Improper Fractions and Mixed Numbers

Sometimes a fraction will take a different form. A fraction with the top number equal to or greater than the bottom number is called an **improper fraction**.

When the numerator is *the same as* the denominator, the fraction *is equal to* 1.

Example 1

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$\frac{4}{4} = 1$ because 4 divided by 4 is 1.



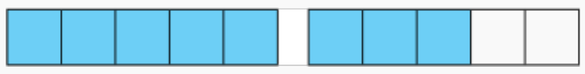
$\frac{4}{4}$ is one whole.

When the numerator *is greater than* the denominator, the fraction *is greater than* 1.

Example 2

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$\frac{8}{5} > 1$ because 8 divided by 5 is 1 with a remainder of 3.



$\frac{8}{5}$ is one whole plus 3 fifths.

A **mixed number** is a "mix" of a whole number and a fraction. An improper fraction can be written as a mixed number using division.

Example 3

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Write $\frac{8}{5}$ as a mixed number.

Step 1

Divide the numerator by the denominator.

$\frac{8}{5} = 5\overline{)8}^{1\ R3}$

Step 2

The quotient becomes the whole number part. (1)

Put the remainder over the original denominator to make the fraction part. $(\frac{3}{5})$

$\frac{8}{5} = 1\frac{3}{5}$ "One and three fifths."

Example 4

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Write $2\frac{1}{4}$ as a fraction.

Step 1

Multiply the whole number by the denominator. Write the product over the denominator of the fraction.

$2 \times 4 = 8$

$\frac{8}{4}$

Step 2

Add the original numerator to the product found in Step 1. (1)

Write the total over the denominator.

$2\frac{1}{4} = \frac{8}{4} + \frac{1}{4} = \frac{9}{4}$

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